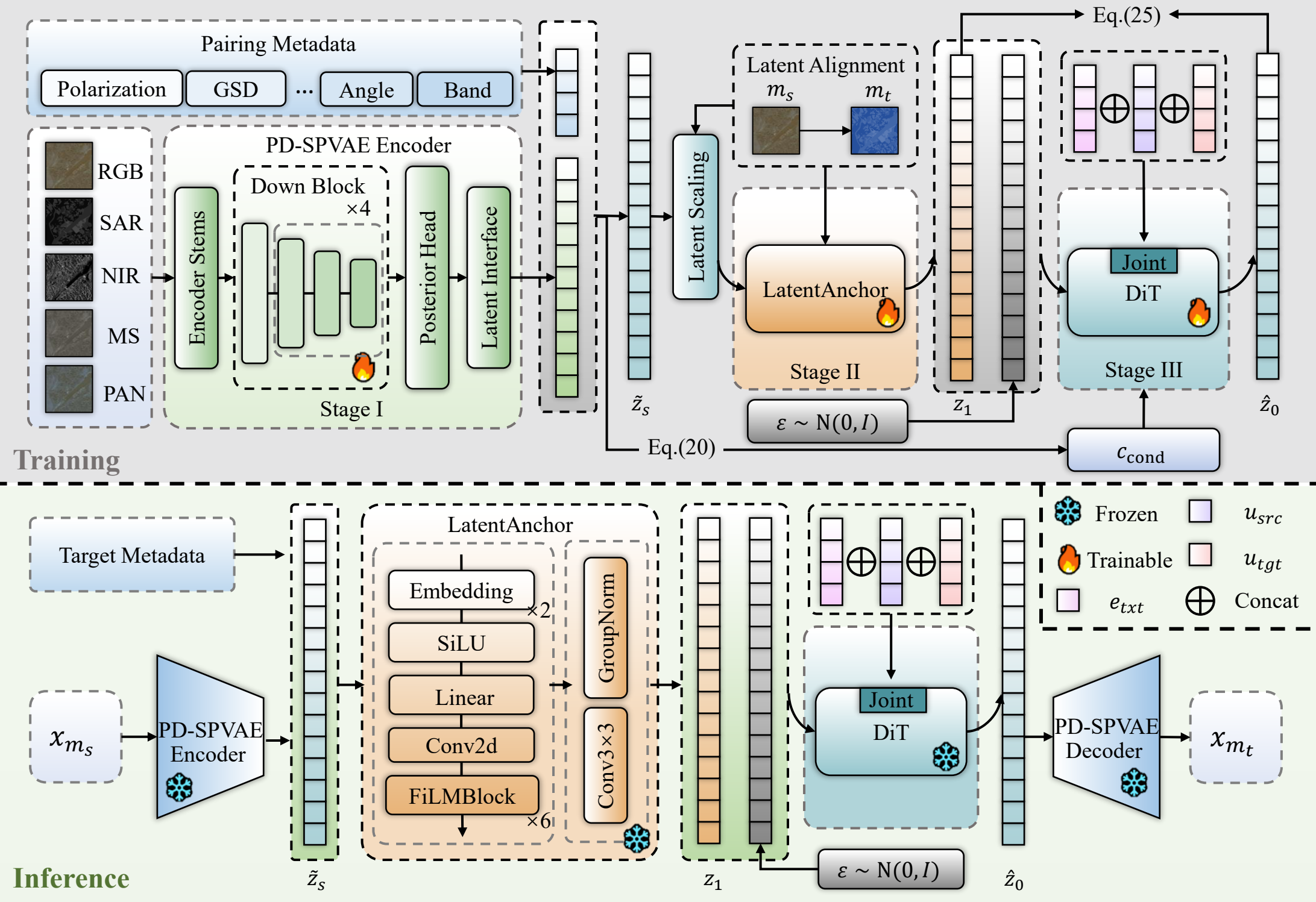

Pairing Metadata
Polarization
GSD
Angle
Band
RGB
SAR
NIR
MS
PAN
PD-SPVAE Encoder
Encoder Stems
Down Block
×4
Posterior Head
Latent Interface
Stage I
Latent Alignment
m_s
m_t
Latent Scaling
LatentAnchor
Stage II
Eq.(25)
Joint
DiT
Stage III
$\tilde{z}_s$
z_1
$\hat{z}_0$
$\varepsilon \sim \mathrm{N}(0, I)$
Eq.(20)
c_{cond}
Training
Target Metadata
x_{m_s}
PD-SPVAE Encoder
LatentAnchor
Embedding
×2
SiLU
Linear
Conv2d
FiLMBlock
×6
GroupNorm
Conv3×3
Frozen
Trainable
e_{txt}
u_{src}
u_{tgt}
Concat
PD-SPVAE Decoder
x_{m_t}
Inference